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Student: Kyle Dick 7206287**Seat Assignment:** Cambridge Test Centre**Testing Date:** July 15, 2025**Start Time:** 5:00 pm**End Time:** 8:00 pm**Test Duration:** 180 minutes**Test Category:****Accommodations:** Medical breaks; Noise reduction headphones; 30 minutes per hour of testing (50% additional test time)**Other accommodations (if applicable):****Test Centre Internal Notes:****Student Preferences:**

INVIGILATOR USE**Invigilator Name:** Jennifer**Actual Start Time:** 5:00 pm**Actual End Time:** 7:04 pm**Notes:**

TEST INFORMATION**Course:** Advanced Topics in Mathematics (MATH73235_101_S1) (MATH73235_101_S1)**Test Name:** test 2**Class Start Time:** 5:00 pm**Class Test Duration:** 120 minutes

AUTHORIZED AIDS**Authorized Aids:** Reference sheet or crib sheet (provide details regarding restrictions) ; Calculator**Specific Instructions:** formula sheet is provided at end of test no other sheets allowed

TEST TYPE DETAILS**Has All Test Information Been Provided (Q9)?:** Yes

Test Type: Paper based

Password Required?: Password:

Software Required:

Answers Recorded On: On Physical Copy of Test

Specific Instructions:

FACULTY INFORMATION

Faculty Contact #:

Classroom:

Preferred Contact Method: email or teams

Preferred Test Retrieval: Scanned and uploaded to Accommodate (within 2 business days)

Faculty Name: Mukto Akash

Faculty Email: makash@conestogac.on.ca

Midterm 2

EXAM CONTENT

MATH73235_Midterm2.pdf

Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): Kyle Dick

Student ID: 7206287

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units** for applied problems.
- **Clearly identify your answer** for each problem.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

Do not write in the table to the right.

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$\iiint_T f(u, v, w) = \int \left(x(u, v, w), y(u, v, w), \frac{\partial(x, y, z)}{\partial(u, v, w)} \right) du dv dw$$

- (b) The main equation for Stokes' Theorem for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \vec{F} \cdot \vec{n} dA \quad \text{Stokes theorem}$$

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

$$\vec{n} = (-2x, -2y, 1) = (2x, 2y, 1)$$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$x^2 + y^2 \hat{i} = r^2 \hat{e}_r$$

$$-2y \hat{j} = -2r \sin \theta \hat{e}_\theta$$

$$\vec{F}_{\text{polar}} = r^2 \hat{e}_r - 2r \sin \theta \hat{e}_\theta$$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$\int_V (\nabla \cdot \vec{F}) dV = \oint \vec{F} \cdot \vec{n} dA$$

$$z^2 = \rho^2 \cos^2 \phi$$

$$r^2 = \rho^2$$

$$(r^2 + z^2) r dr d\theta dz$$

$$\therefore \vec{\nabla} \cdot \vec{F}_{\text{spherical}} = \rho^2 (1 + \cos^2 \phi)$$

$$= r^2(1) + z^2.$$

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$$f(r, \theta, z) = r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2$$

$$1 - \cos(2t) = \sin^2(t)$$

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$$1 + \cos(2t) = \cos^2(t)$$

2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function μ

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

$$h_1 = h_r = 1; \quad h_2 = h_\theta = r; \quad h_3 = h_z = 1$$

$$x = r \cos(\theta)$$

$$y = r \sin \theta$$

$$z = z$$

$$\nabla^2 f_{\text{cyl.}} = \frac{1}{r} \left(\frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} (r \cos \theta)^2 \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial}{\partial \theta} (r \cos \theta)^2 \right) + \frac{\partial}{\partial z} \left(\frac{\partial}{\partial z} (r \cos \theta)^2 \right) \right)$$

$$\frac{d}{d\theta} (\cos \theta - \sin \theta) = -\sin \theta - \cos \theta$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (r (2r \cos^2 \theta + 2r \sin^2 \theta + 0)) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} (r^2 (2) (-\sin \theta) + r^2 \cos \theta (2)) \right) + \frac{\partial}{\partial z} (r (2z)) \right]$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial \theta} (2r \cos \theta - 2r \sin \theta) + 2r \right]$$

$$= \frac{2r}{r} [1 - \sin \theta - \cos \theta + 1] = 2 - \sin \theta - \cos \theta = \nabla^2 F_{\text{cyl.}} \quad (\text{Go to page 9 for } \nabla^2 F_{\text{cyl.}} \text{ soln})$$

$$f(\rho, \theta, \phi) = (\rho^2 \cos^2 \theta \sin^2 \phi) + \rho^2 \sin^2 \theta \sin^2 \phi + \rho^2 \cos^2 \phi$$

$$\nabla^2 F_{\text{sphere}}$$

$$= \rho^2 (\cos^2 \theta \sin^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \phi)$$

$$\sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \cos^2 \phi$$

$$= \rho^2.$$

$$= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\frac{\rho^2 \sin \phi}{1} \cdot \frac{\partial}{\partial \rho} (\rho^2) \right) \right]$$

q_2 & $q_3 = \text{zero} \therefore f$ is only in terms of ρ .

$$\text{Aside: } \left(\oint_S (\nabla \times F) \cdot \vec{n} dA = \oint_C F \cdot d\vec{r} \right) = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} 2\rho^3 \sin \phi \right] = \frac{6\rho^2 \sin \phi}{\rho^2 \sin \phi} = \frac{6\rho^2}{\rho^2} = 6$$

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3. (10 points) Use Spherical coordinates to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$. $\square$

Solution.

$$\int_0^{2\pi} \int_0^2 \rho \sin \phi \, d\rho \, d\theta, \quad \phi = \pi/6$$

$$\rho \sin \phi \, d\rho \, d\theta.$$

$$= \int_0^2 \rho \, d\rho \int_0^{2\pi} d\theta \cdot \sin(\pi/6)$$

$$= \left. \frac{\rho^2}{2} \right|_0^2 \left. \theta \right|_0^{2\pi} \cdot (0.5) = \frac{(2)(2\pi)}{2} = 2\pi.$$



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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:
- $\int_S \vec{F} \cdot \hat{n} \, dA$

Solve in spherical.

$$\int_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\int_T (\nabla \cdot \vec{F}) \, dV,$$

where $F = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

for volume.

Solution. $\vec{F} = (x, y, z)$ $S: x^2 + y^2 + z^2 = 4$; in spherical: $\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

$$\vec{F} \cdot \hat{n} = \rho. \quad \vec{F} \cdot \hat{n} = \rho\hat{e}_\rho \cdot \hat{e}_\rho = \rho = 2.$$

$$\int_S \vec{F} \cdot \hat{n} \, dA = \int_0^{2\pi} \int_0^\pi \int_0^2 2 \, d\rho \, d\phi \, d\theta = (2)(2)(2\pi) = 8\pi$$

$$\int_T (\nabla \cdot \vec{F}) \, dV = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x & y & z \end{vmatrix} = (0, 0, 0) \quad \nabla \cdot \vec{F} = \partial_x + \partial_y + \partial_z$$

$$\vec{F} = \rho\hat{e}_\rho, \quad \nabla \cdot \vec{F} = \begin{vmatrix} \hat{e}_\rho & \hat{e}_\phi & \hat{e}_\theta \\ \partial_\rho & \partial_\phi & \partial_\theta \\ \rho & 0 & 0 \end{vmatrix} = (0, 0, 0) ?$$

$$\int_0^{2\pi} \int_0^\pi \int_0^2 0 \, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 0$$

□

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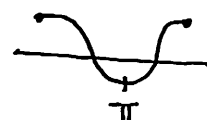
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Continue problem 4

Solution.

Don't confuse curl & divergence

$$\vec{\nabla} \cdot \vec{F} = 1 + 1 + 1 = 3. \quad \rho = 2.$$



$$\int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} \int_{\rho=0}^2 3 \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi = \left. +2\rho^3 \right|_0^2 \left. (-\cos \phi) \right|_0^{\pi} = 12(2)(2\pi)(-1 - 1)$$

$$\begin{aligned} \left. \frac{2}{3} \rho^3 \right|_0^2 &= 8(2\pi)(-\cos \phi) \Big|_0^{\pi} \\ &= 8(2\pi)(-(-1) + 1) \\ &= 8(2\pi)(2) \rightarrow 0? \end{aligned}$$

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Q1.

5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$\frac{\partial(x, y)}{\partial(u, v)} = |J| = |h_u h_v| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} ve^u & -ve^{-u} \\ e^u & e^{-u} \end{vmatrix}$$

$$= ve^u - e^{-u} - (-ve^{-u}e^u) = ve^u - e^{-u} + ve^{-u} - e^u$$

$$= ve^u - e^u + ve^{-u} - e^{-u} = (v-1)e^u + (v-1)e^{-u} = (v-1)(e^u - e^{-u})$$

$$= \int_{v=0}^u \int_0^u (v-1)(e^u - e^{-u}) du dv$$

$$= \left(\frac{v^2}{2} - v \right) \bigg|_0^u (e^u - e^{-u})$$

$$= \left(\frac{u^2}{2} - u \right) (e^u - e^{-u})$$

$$= \int_{u=0}^3 \left(\frac{u^2}{2} - u \right) (e^u - e^{-u}) du = \int_0^3 u dv = uv - \int v du.$$

$$= \left(\frac{u^2}{2} - u \right) (2e^u) \bigg|_0^3 - \int_0^3 (2e^u)(u-1) du$$

$$a^b - a^{-b}$$

$$a^b(1 - a^{-2b}) = a^b - a^{-b}$$

$$a^b - \frac{1}{a^b} \neq a^2$$

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Extra Page: Must submit even if unused.

$$\int_0^3 (2e^u)(u-1)$$

$$du = 1 \quad u = u-1$$

$$dv = 2e^u \quad v = 2e^u$$

$$= uv - \int v du = (u-1)(2e^u) \Big|_0^3 - \int_0^3 (2e^u)(1)$$

$$= (\text{From prev. page}) : \left(\frac{u^2}{2} - u \right) (2e^u) \Big|_0^3 - (u-1) \Big|_0^3 (2e^u) \Big|_0^3$$

$$- 2e^u \Big|_0^3$$

~~2e^0~~

$$= \left(\frac{9}{2} - 3 \right) (2e^3) - (3-1) 2e^3 - 2e^3$$

$$= 2e^3 \left(\frac{9}{2} - 3 - 2 \right) - 2e^3$$

$$= 2e^3 \left(\frac{9}{2} - 3 - 2 \right) - 2e^3$$

$$= 2e^3 - 2e^3$$

NOTE: May have missed $f(x,y) \Rightarrow \underline{f(u,v)}$ in $\iint_{u,v} f(u,v) h_u h_v du dv$.

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Q2. p3

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$$\nabla^2 F = r^2 + z^2.$$

$$= \frac{1}{r} \left[\frac{d}{dr} (r(2r)) + \frac{\partial}{\partial z} (r^2 z) \right] = \frac{1}{r} [4r + 2r] = \underline{\underline{6}}.$$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$